

CIE

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TITLE: Smart AI Cataloging Application

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ABSTRACT

1.1 Problem Background

Traditional inventory management and e-commerce workflows suffer from manual cataloging bottlenecks, where products must be categorized and tagged by hand. This process is highly prone to human error, inconsistent naming conventions, and significant delays as inventory scales. Businesses often struggle with "data silos" where unstructured product images and descriptions are difficult to search or filter, leading to poor user experiences and inefficient stock tracking. Without an automated system, the time-to-market for new products is slow, and maintaining a clean, organized catalog becomes an expensive operational burden.

1.2 Objective of the Project

The primary objective is to develop a "Smart Cataloging Application" that automates the organization of inventory. By utilizing Artificial Intelligence, the system aims to analyze raw product data—such as images and text—to automatically assign categories, generate descriptive tags, and optimize searchability, ensuring a seamless flow from product intake to a structured digital storefront.

1.3 Proposed Solution / System Overview

The proposed system is a full-stack web application built on **Next.js**, integrating Computer Vision and Natural Language Processing (NLP) models to handle automated classification. The architecture follows a serverless approach where users upload product details; the backend then routes these to an AI engine (such as OpenAI's GPT-4o or specialized Vision models) to extract attributes. The system employs a "Smart Suggestion" UI, allowing users to review AI-generated labels before they are committed to the central database, ensuring both speed and high data integrity.

1.4 Key Features or Functionality

Key functionalities include **AI-Powered Image Recognition** for automatic category assignment and **Attribute Extraction** (e.g., color, material, brand) from unstructured descriptions. The application features a **Real-Time Inventory Dashboard** built with Next.js for high-speed performance, a bulk-upload utility for rapid cataloging, and a dynamic search engine that utilizes AI-generated tags. Additionally, the system includes an "Edit & Confirm" loop to maintain human-in-the-loop accuracy for edge-case products.

1.5 Technology / Method Used

The project utilizes the **MERN or T3 Stack** (Next.js, Tailwind CSS, and a robust database like PostgreSQL or MongoDB). **Next.js API Routes** serve as the bridge between the frontend and

AI inference models. For image analysis, the system integrates **OpenAI Vision API** or **TensorFlow.js**. The development methodology follows an Agile approach: starting with the core classification engine, followed by the integration of the Next.js frontend, and concluding with performance optimization through edge caching and database indexing.

1.6 Expected Outcome & Benefits

The expected outcome is an intelligent, responsive web application that reduces product cataloging time by up to 80%. Key benefits include **unmatched consistency** in inventory tagging, a **drastic reduction in operational costs**, and **enhanced searchability** for end-users. By automating the "boring" parts of data entry, businesses can scale their inventory rapidly without a proportional increase in headcount, while the modern Next.js framework ensures a fast, SEO-friendly interface.

1.7 Original Contribution Statement

This project contributes a specialized integration of Generative AI with Next.js specifically tailored for inventory logistics. It bridges the gap between raw unstructured data and organized databases, providing a blueprint for AI-native retail management systems.